



UW/CS/F25/CS201/A3

Department of Computer Science

Assignment 3

Course Code: CS-201

Course Title: Data Structures

Total Marks: 20

Date: 16-01-2025

Name: _____

Registration No. _____

Discipline/Semester: BSCS- 3rd A

Submission deadline: 19-01-2026

Instructions:

- Late submissions will not be entertained.
- Copied assignments will be marked ZERO.
- Use of AI tool is strictly not allowed.
- Submit the assignment in hard form on A4 pages.

CLO3, GA3, C4

QUESTION

Build an AVL Tree using the given data and write pseudocode to perform various operations on the tree, including its creation, insertion of nodes, and deletion for specific values. Ensure that the AVL Tree maintains its self-balancing property during all operations. **Analyze** the time complexity of each operation and explain how the balancing mechanism of the AVL Tree enhances its efficiency compared to a standard binary search tree.

Data: 15, 10, 20, 8, 12, 17, 25, 6, 11

Note: Show all steps for the creation of AVL Tree.